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Письмо редактору

Судьба Земли во время гигантских фаз Солнца

Новые данные, полученные в результате приливного моделирования ab initio и изучения потери массы на асимптотической ветви гигантов

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Аннотация

Контекст. Долгосрочная эволюция планетных систем вокруг звезд солнечного типа зависит от взаимодействия между расширением звезды, приливными взаимодействиями и потерей массы на этапах ветви красных гигантов (ВКГ) и асимптотической ветви гигантов (АВГ). Однако эффективность приливной диссипации и скорость потери массы на АВГ до сих пор изучены недостаточно, что приводит к значительной неопределенности в прогнозировании судьбы планетных систем, в частности Земли, вращающейся вокруг стареющего Солнца.

Цели. Мы заново оцениваем вероятность выживания Земли и планет внутренней Солнечной системы в период после главной последовательности эволюции Солнца, уделяя особое внимание влиянию обновленных моделей приливной диссипации и различных показателей потери массы на асимптотической ветви гигантов.

Методы. Мы смоделировали эволюцию орбиты Земли, используя треки звездной эволюции для звезды с массой Солнца, в которых учтены последние данные о приливном затухании, полученные в результате расчетов ab initio. Мы сравнили эти результаты с результатами, полученными с использованием ранее опубликованных и общепринятых моделей приливного затухания, а также изучили различные показатели потери массы на асимптотической ветви гигантов.

Результаты. Мы обнаружили, что прогнозируемая судьба Земли сильно зависит от модели приливного взаимодействия и предполагаемой скорости потери массы. Согласно обновленным данным о приливном взаимодействии, Земля переживет фазы красного гиганта и асимптотической ветви гигантов на Солнце. Однако использование более ранних данных о приливном взаимодействии приводит к поглощению Земли на этапе асимптотической ветви гигантов. Кроме того, поглощение происходит при низкой скорости потери массы на этапе асимптотической ветви гигантов, и наоборот. Если использовать наблюдаемую скорость потери массы звездой на асимптотической ветви гигантов L_2 в созвездии Парусов в качестве косвенного показателя будущей скорости потери массы Солнца на этой стадии, то, согласно нашей оценке приливной диссипации, Земля переживёт эту фазу.

Выводы. В настоящее время вопрос о том, выживет ли Земля и внутренняя часть Солнечной системы, остается открытым и в значительной степени зависит от того, как будут учитываться приливная диссипация и потеря массы звездой. Учитывая существующие погрешности наблюдений в отношении скорости потери массы на асимптотической ветви гигантов, дальнейшая судьба Земли остается неопределенной, что подчеркивает необходимость более точных данных о поздних стадиях звездной эволюции. Однако, учитывая наблюдаемые характеристики Солнца на этапе асимптотической ветви гигантов, можно предположить, что Земля переживет фазу красного гиганта.

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Ключевые слова: методы: численные / взаимодействие Земли / планет и звезд / двойные звезды: тесные / звезды: эволюция / планетные системы

1. Введение

Когда в ядре звезды заканчивается водород, она сходит с главной последовательности (ГП) и расширяется на протяжении фаз ветви красных гигантов (ВКГ) и асимптотической ветви гигантов (АВГ) (см., например, Kippenhahn & Weigert 1990). В ходе этой эволюции звезда увеличивается в размерах и теряет массу из-за звездного ветра. В случае с Солнцем это расширение приведет к тому, что звезда станет больше нынешней орбиты Земли (1,2–1,5 $M_{\odot}$ в зависимости от скорости потери массы на асимптотической ветви гигантов). Это окажет значительное влияние на эволюцию орбит планет Солнечной системы. Судьба внутренних планет Солнечной системы, и в частности Земли, является предметом дискуссий в научной литературе, и разные исследования приходят к разным выводам. Некоторые исследования предполагали, что Земля будет поглощена расширяющимся Солнцем во время RGB (например, Рыбицки и Денис 2001; Шредер и Смит 2008; Ланца и др. 2023), в то время как другие предполагали, что Земля переживет фазу RGB (например, Сакманн и др. 1993; Расио и др. 1996; Нордхаус и др. 2010; Го и др. 2016). В этих исследованиях часто использовались разные предположения о скорости потери массы и приливных взаимодействиях, что приводило к разным выводам. Кроме того, в некоторых исследованиях рассматривалось влияние взаимодействия планет на эволюцию орбит во внутренней части Солнечной системы (например, Mogavero & Laskar 2021), но при этом не учитывалась структурная эволюция Солнца. Наблюдения показывают, что вокруг белых карликов могут вращаться планеты земного типа (например, Чжан и др., 2024), что свидетельствует о возможности существования планет, подобных Земле, на стадиях красного гиганта и асимптотической ветви гигантов. Однако точная судьба Земли и внутренней части

Солнечной системы до сих пор неясна и требует дальнейшего изучения.

Влияние потери массы на эволюцию орбит планет Солнечной системы впервые было изучено Сакманном и др. (1993), которые исходили из предположения о сохранении углового момента при потере массы звездой (что фактически приводит к расширению орбиты). Рыбицкий и Денис (2001) проанализировали не только потерю массы, но и другие факторы, такие как приливные взаимодействия, аккреция массы за счёт звёздного ветра, испарение под воздействием звёздного излучения и сопротивление ветра. Авторы пришли к выводу, что на эволюцию орбит в большей степени влияют потеря массы и приливные взаимодействия. Шрёдер и Смит (2008) и Нордхаус и др. (2010) объединили данные о потере массы и приливных взаимодействиях и пришли к разным выводам (поглощение для Шрёдера и Смита, 2008 и выживание для Нордхауса и др., 2010). Различия в выводах этих исследований могут быть связаны с разными предположениями о скорости потери массы во время красного гиганта, где Нордхаус и др. (2010) предполагали более высокую скорость потери массы, чем Шрёдер и Смит (2008). Эта зависимость была дополнительно проанализирована Го и др. (2016), которые провели детальное исследование влияния различных скоростей потери массы во время красного гиганта на эволюцию орбиты. При низкой скорости потери массы во время фазы красного гиганта звезда не только проводит в этой фазе больше времени, но и, что более важно, значительно увеличивается в размерах. Это увеличивает продолжительность фазы и приводит к более сильным приливным взаимодействиям, которые уменьшают орбиту Земли, что в конечном итоге может привести к поглощению планеты звездой, и наоборот. Основываясь на данных о скорости потери массы во время фазы красного гиганта, полученных в результате наблюдений Макдональд и Зейлстра (2015), Го и др. (2016) пришли к выводу, что Земля переживёт фазу красного гиганта. Ланца и др. (2023) учли влияние остаточного эксцентриситета — стохастического процесса, при котором колебания плотности вокруг планеты приводят к колебаниям эксцентриситета примерно на 10^{-5} во время средней стадии, которые усиливаются до порядка 10^{-2} во время поздней стадии, что делает выживание Земли стохастическим процессом.

These previous studies either neglected the effect of tidal interactions altogether (e.g. Sackmann et al. 1993) or used simplified prescriptions for the tidal interactions based on the equilibrium tide theory of Zahn

(1966, 1977) (e.g. Rybicki & Denis 2001; Schröder & Smith 2008; Guo et al. 2016; Lanza et al. 2023). These prescriptions are based on the assumption that the tidal dissipation is dominated by the equilibrium tide and use parametrisations for stellar properties. However, recent studies have improved these prescriptions by using ab initio modelling of the tidal dissipation based on the latest understanding of the internal structure and dynamics of evolved stars and including dynamical tides associated with progressive internal gravity waves (see Esseldeurs et al. 2024, and references therein). Along the same lines, mass loss during the AGB phase is also still uncertain (Decin 2021) and can affect the orbital evolution of planetary or binary systems significantly.

We aim to improve the modelling of the tidal interactions by using the prescriptions of Esseldeurs et al. (2024) for the tidal dissipation. We also use the tools developed in Esseldeurs et al. (2026) to model the orbital evolution, which allows us to take the tidal interactions and the changes through stellar winds into account. Using these tools, we model the orbital evolution of the inner Solar System planets during the Sun's RGB and AGB phases and determine their fate during these phases.

In Sect. 2 we describe the tools we used to model the orbital evolution of the inner Solar System. In Sect. 3 we outline the stellar evolutionary models we used for the Sun. In Sect. 4 we present the results of the orbital evolution of the inner Solar System and discuss the effect of different tidal dissipation prescriptions and AGB mass-loss rates on the fate of the Earth.

2. Modelling the orbital evolution

To model the orbital evolution of the inner Solar System, we used the tools developed in Esseldeurs et al. (2026). To follow changes in planetary orbital parameters, we considered tidal interactions in the Sun as well as stellar winds,

$$\frac{1}{a}\frac{\mathrm{d}a}{\mathrm{d}t} = \frac{1}{a}\frac{\mathrm{d}a}{\mathrm{d}t}\bigg|_{\mathrm{wind}} + \frac{1}{a}\frac{\mathrm{d}a}{\mathrm{d}t}\bigg|_{\mathrm{tides}} . \tag{1}$$

For all simulations, the eccentricity remained negligible, but was taken into account for consistency. The equations for their evolution are given in Appendix A. These differential equations were integrated for each planet independently, and we did not take the effect of planet-planet interactions into account. Earth only crosses the 5:3 mean motion resonance with Venus during the RGB phase, but this is expected to have only a minor effect on the orbital evolution of Earth (see Appendix D).

The changes through stellar winds are given by

$$\left\langle\frac{1}{a}\frac{\mathrm{d}a}{\mathrm{d}t}\right\rangle\bigg|_{\mathrm{wind}} = -2\frac{\dot{m}_1}{m_1}\left(1-\frac{\beta}{q}-\eta(1-\beta)\frac{1+q}{q}-\frac{1-\beta}{2}\frac{1}{1+q}\right), \tag{2}$$

where $\langle\rangle$ denotes an orbit-averaged quantity, q is the mass fraction, η is the specific angular momentum of material lost in units of the orbital angular momentum of the system per reduced mass, and $\beta = \frac{1+e^2}{(1-e^2)^{3/2}}\beta_c$ is the enhanced mass-accretion efficiency effect accounting for the change in the orbital separation for eccentric orbits. η and β_c were calculated using the analytical prescriptions proposed by Saladino et al. (2019).

The changes through tides are given by

$$\frac{1}{a}\frac{da}{dt}\Big|_{\text{tides}} = -\frac{5}{4\pi}\frac{m_2}{m_1}\left(\frac{R_\star}{a}\right)^5 \sum_{m=0}^{\infty} \sum_{n=-\infty}^{\infty} n\Omega_o\, |A_{2,m,n}(e)|^2\, I(k_n^{2,m}), \tag{3}$$

where $A_{2,m,n}(e)$ are the tidal coefficients (see Ogilvie 2014 Table 1 and Esseldeurs et al. 2026 for details), and $\Im(k_n{}^{2,m})$ are the imaginary parts of the tidal Love numbers, which were calculated using the prescriptions of Esseldeurs et al. (2024). These equations assume a negligible stellar rotation, which is a good approximation for evolved stars that have not been spun up through tidal interactions. This is a valid approximation for our Solar System calculations since tidal interactions are too weak to significantly spin the Sun up (Madappatt et al. 2016). The simulation either ended at the end of the stellar evolution model (a cool WD with $L = 10^{-1}$ $L_\odot$) or when the planet was engulfed by the star (when the orbital separation was smaller than the Roche radius).

3. Stellar evolutionary models

We used the stellar evolutionary code MESA (Paxton et al. 2011, 2013, 2015, 2018, 2019; Jermyn et al. 2023) to compute the evolution of the Sun from the pre-main sequence to the white dwarf phase. We used the same input physics as in Esseldeurs et al. (2024), which were based on Cinquegrana & Joyce (2022) to be consistent with the Monash code (Karakas & Lattanzio 2007) during the evolved phases. The initial mass of the Sun was set to 1 $M_\odot$, and we used a metallicity of $Z = 0.0134$ (Asplund et al. 2009). For the mass-loss rate during the RGB phase, we used the Reimers mass-loss rate prescription (Reimers 1975) with $\eta_{\text{Reimer}} = 0.477$, as calibrated from observations by McDonald & Zijlstra (2015), and we used the Blöcker mass-loss rate prescription (Blöcker 1995) with $\eta_{\text{Blöcker}}$ between 0.01 and 0.5 (0.1 in Esseldeurs et al. 2024) for the mass-loss rate during the AGB phase, where 0.05 was taken as a reference value. A Kippenhahn diagram is shown in Appendix B.

4. Orbital evolution predictions

Using the stellar and orbital evolution models described in the previous section, we predicted the orbital evolution of the inner Solar System planets from the PMS until the end of the WD phase. The results are shown in Fig. 1, where we show the evolution of the orbital separation of the planets as a function of time until the WD phase. During the RGB phase, the orbital separation of the planets increases due to the mass loss of the star, where the increase is larger for planets that are farther away from the star. Mercury and Venus are not able to move outward fast enough to avoid engulfment and are engulfed during the RGB phase. Earth and Mars are able to move outward fast enough to avoid engulfment during the RGB phase. During the HB phase, the orbital separation of the planets stabilises as the star contracts, while during the AGB phase, the orbital separation of the planets increases again due to the mass loss of the star. During this phase, Earth and Mars are able to move outward fast enough to avoid engulfment. During the WD phase, the orbital separation of the planets stabilises again as the star contracts.

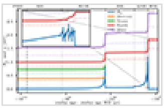

Fig. 1.

Orbital evolution (solid line, and the corresponding Solar Roche lobe radius, dotted line) of the inner Solar System during the evolved phases of the Sun. The different lines correspond to different planets (Mercury in orange, Venus in green, Earth in red, and Mars in purple). The current age of the Sun is indicated by the vertical dashed line.

4.1. Impact of tidal dissipation modelling

To understand the effect of the different tidal dissipation prescriptions on the orbital evolution, we compared the results using the tidal prescriptions of Esseldeurs et al. (2024) to a Zahn (1966)-based prescription as used by Mustill & Villaver (2012) (see Appendix C for a comparisson). The main difference between the prescriptions is the treatment of the frequency dependence of the turbulent viscosity. At Earth's orbit, the difference between the viscosity prescriptions is largest, making the choice of prescription crucial for Earth's orbital evolution.

The results are shown in Fig. 2. The older prescriptions of Mustill & Villaver (2012) predict stronger tidal dissipation, which leads to a reduction in the orbital widening, and thus, to Earth leaving the RGB phase on a much closer orbit. Earth therefore starts at a shorter orbital distance at the start of the AGB phase, resulting in engulfment during the AGB phase. In contrast, our models predict Earth to be surviving the Sun's RGB and subsequent AGB evolution, assuming $\eta_{\text{Blöcker}} = 0.05$. As Earth is expected to move farther outward compared to previous studies, the stochastic eccentricity fluctuations induced by Lanza et al. (2023) are less important for Earth.

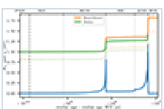

Fig. 2.

Orbital evolution of the Earth during the evolved phases of the Sun for different tidal dissipation prescriptions. In orange, we plot Esseldeurs et al. (2024) and in green, we plot Mustill & Villaver (2012) (based on Zahn 1966; see Appendix C).

Observationally, the tidal dissipation was tested in evolved stars using the circularisation of binary systems. In the RGB phase, Beck et al. (2018, 2022, 2024), and Dewberry & Wu (2025) have shown that the equilibrium tide dominates the tidal dissipation, but that the observed circularisation is stronger than predicted by the equilibrium tide using the Duguid et al. (2020) viscosity. In the AGB phase, Esseldeurs et al. (2026) have shown that the observed circularity of the binary system π^1 Gru is unexpected given current tidal predictions. This suggests that tidal dissipation in evolved stars might be stronger than predicted by current models, making Earth's engulfment more likely. However, these observations are based on binary circularisation, which involves higher-frequency tides than circular migration of orbits such as in the Earth-Sun system, and it might therefore not be directly applicable.

4.2. Impact of mass-loss rate prescriptions

In the past, the importance of mass loss during the RGB and AGB phases was highlighted by several studies (e.g. Sackmann et al. 1993; Guo et al. 2016). Guo et al. (2016) investigated the effect of different η_{Reimers} values (for describing the mass-loss rate during the RGB phase) on the orbital evolution and found that for values below $\eta_{\mathrm{Reimers}} = 0.46$, Earth is engulfed during the RGB phase, and vice versa. As our tidal dissipation is weaker, this value will decrease when recomputed using our model. Because η_{Reimers} is relatively well constrained by observations as $\eta_{\mathrm{Reimer}} = 0.477 \pm 0.07$ (McDonald & Zijlstra 2015), the fate of the Earth during the RGB phase is relatively well constrained, and the Earth is expected to survive the RGB phase, but the engulfment is still within the uncertainty.

The AGB mass-loss rates are more uncertain. Different prescriptions predict mass-loss rates that differ by more than one order of magnitude (Decin 2021). Additionally, within the same prescription, different values of the free parameter $\eta_{\mathrm{Blöcker}}$ are used in the literature. To constrain this parameter observationally for our Sun, we considered the observations of the AGB star L_2 Pup, which is an AGB star with an initial mass close to the Sun's (0.98 $M_\odot$; Kervella et al. 2016) and can thus be used as a proxy for the Sun during the AGB phase. L_2 Pup is surrounded by a dusty disk, inside which a potential planet of mass 12 ± 16 M_{Jup} was detected (Kervella et al. 2016). There have been two different estimates of the mass-loss rate of L_2 Pup, one estimate based on the dust emission, which gives a mass-loss rate of $\dot{m}_1 = 1 \times 10^{-6}$ $M_\odot$ yr^{-1} (Haworth et al. 2018), and the other estimate based on the CO emission, which gives a much lower mass-loss rate of $\dot{m}_1 = 1.2 \times 10^{-8}$ $M_\odot$ yr^{-1} (Hoai et al. 2022). We refer to Appendix E for a discussion, but the large difference between the two estimates already highlights the uncertainty in the AGB mass-loss rates. With the Blöcker mass-loss rate prescription, the dust-based mass-loss rate corresponds to $\eta_{\text{Blöcker}} = 2.5$, while the CO-based mass-loss rate corresponds to $\eta_{\text{Blöcker}} = 0.03$. Values between 0.01 and 1 are commonly used in the literature. We took $\eta_{\text{Blöcker}} = 0.05$ as a reference value (see Appendix E), but we also explored the effect of different values between 0.01 and 0.5 for $\eta_{\text{Blöcker}}$ on the orbital evolution of the Earth. It is worth noting that other AGB mass-loss prescriptions exist, such as the period-based prescription of Vassiliadis & Wood (1993), which cannot be directly compared to the luminosity-based Blöcker prescription. However, $\eta_{\text{Blöcker}}$ can be calibrated to reproduce a similar mass-loss evolution within the explored parameter space.

The orbital evolution of Earth during the thermally pulsing AGB phase for different $\eta_{\text{Blöcker}}$ values is shown in Fig. 3. For low $\eta_{\text{Blöcker}}$ values (0.01 and 0.02), the Sun's radius exceeds the Roche lobe radius. Under normal circumstances, this would lead to Earth's engulfment. However, this occurs during a thermal pulse lasting only a few hundred years, making it unclear whether the interaction is strong enough to cause engulfment. For $\eta_{\text{Blöcker}} = 0.01$, the Solar radius grows nearly as large as Earth's orbit, making engulfment likely. For $\eta_{\text{Blöcker}} = 0.02$ (and 0.03, not shown), the Solar radius exceeds the Roche lobe radius only slightly (filling factor of 7%), making the outcome uncertain. To anccurately model this would require stellar evolution calculations including the star-planet interaction during the thermally pulsing AGB phase (e.g. MESA in BINARY mode; Paxton et al. 2015), combined with an analytical model of the orbital evolution due to mass transfer (e.g. Parkosidis et al. 2026a,b), which is beyond the scope of this paper. For $\eta_{\text{Blöcker}} \geq 0.04$, the Solar radius remains below the Roche lobe radius, allowing Earth to survive the AGB phase. Because the AGB mass-loss rates are uncertain, Earth's fate remains uncertain, although the observational values of L_2 Pup suggest that Earth will likely survive the AGB phase.

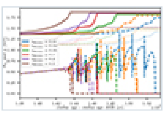

Fig. 3.

Orbital evolution of the Earth during the evolved phases of the Sun for different AGB mass-loss rates. The different lines correspond to different values for $\eta_{\text{Blöcker}}$. The dashed lines correspond to the Solar radius, and the dotted lines correspond to the Solar Roche lobe radius. After the radius of the Sun exceeds the Roche lobe radius, the solid line for the orbital distance is changed to a dashed line to highlight the uncertainty in the model after this point (see text).

5. Conclusion

Our results show that the fate of Earth and the inner Solar System during the Sun's evolved phases strongly depends on tidal interactions and AGB mass-loss rates. Using updated tidal dissipation prescriptions based on ab initio modelling of evolved stars, we found that Earth survives the RGB and AGB phases. In contrast, previous tidal prescriptions predicted engulfment during the AGB phase. We also found that low AGB mass-loss rates lead to engulfment, while high rates allow Earth to survive. Since the AGB mass-loss rates remain observationally uncertain, the final fate of Earth is still unclear. However, the observed properties of L_2 Pup, which we used as a proxy for the future Sun, suggest that Earth will likely survive the AGB phase. Future observational constraints, combining spatially resolved interferometric data with the most recent hydrochemical simulations, are needed to better constrain AGB mass-loss rates and hence the fate of the inner Solar System. Additionally, the number of detected planets around red giants is expected to increase substantially in the coming years, particularly with the PLATO mission (Rauer et al. 2025). This will enable us to conduct population studies of the planetary orbital evolution around evolved stars and help us to constrain the future evolution of the Earth-Sun system.

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References

1. Asplund, M., Grevesse, N., Sauval, A. J., & Scott, P. 2009, ARA&A, 47, 481
[NASA ADS] [CrossRef] [Google Scholar]
2. Beck, P. G., Mathis, S., Gallet, F., et al. 2018, MNRAS, 479, L123
[Google Scholar]

3. Beck, P. G., Mathur, S., Hambleton, K., et al. 2022, A&A, 667, A31 [NASA ADS] [CrossRef] [EDP Sciences] [Google Scholar]
4. Beck, P. G., Grossmann, D. H., Steinwender, L., et al. 2024, A&A, 682, A7 [NASA ADS] [CrossRef] [EDP Sciences] [Google Scholar]
5. Bedding, T. R., Zijlstra, A. A., Jones, A., et al. 2002, MNRAS, 337, 79 [NASA ADS] [CrossRef] [Google Scholar]
6. Blöcker, T. 1995, A&A, 297, 727 [NASA ADS] [Google Scholar]
7. Cinquegrana, G. C., & Joyce, M. 2022, Res. Notes AAS, 6, 77 [Google Scholar]
8. Danilovich, T., Teyssier, D., Justtanont, K., et al. 2015, A&A, 581, A60 [NASA ADS] [CrossRef] [EDP Sciences] [Google Scholar]
9. Decin, L. 2021, ARA&A, 59, 337 [NASA ADS] [CrossRef] [Google Scholar]
10. Dewberry, J. W., & Wu, Y. 2025, ApJ, 984, 137 [Google Scholar]
11. Duguid, C. D., Barker, A. J., & Jones, C. A. 2020, MNRAS, 497, 3400 [Google Scholar]
12. Esseldeurs, M., Mathis, S., & Decin, L. 2024, A&A, 690, A266 [NASA ADS] [CrossRef] [EDP Sciences] [Google Scholar]
13. Esseldeurs, M., Decin, L., De Ridder, J., et al. 2026, Nat. Astron., 10, 124 [Google Scholar]
14. Goldreich, P., & Keeley, D. A. 1977, ApJ, 212, 243 [NASA ADS] [CrossRef] [Google Scholar]
15. Guo, J., Lin, L., Bai, C., & Liu, J. 2016, Ap&SS, 361, 122 [NASA ADS] [CrossRef] [Google Scholar]
16. Hanasoge, S., Gizon, L., & Sreenivasan, K. R. 2016, Ann. Rev. Fluid Mech., 48, 191 [Google Scholar]
17. Haworth, T. J., Booth, R. A., Homan, W., et al. 2018, MNRAS, 473, 317 [Google Scholar]
18. Hoai, D. T., Nhung, P. T., Darriulat, P., et al. 2022, MNRAS, 510, 2363 [NASA ADS] [CrossRef] [Google Scholar]
19. Hurley, J. R., Tout, C. A., & Pols, O. R. 2002, MNRAS, 329, 897 [Google Scholar]
20. Hut, P. 1981, A&A, 99, 126 [NASA ADS] [Google Scholar]
21. Jermyn, A. S., Bauer, E. B., Schwab, J., et al. 2023, ApJS, 265, 15 [NASA ADS] [CrossRef] [Google Scholar]
22. Jura, M., Chen, C., & Plavchan, P. 2002, ApJ, 569, 964 [NASA ADS] [CrossRef] [Google Scholar]
23. Karakas, A., & Lattanzio, J. C. 2007, PASA, 24, 103 [NASA ADS] [CrossRef] [Google Scholar]
24. Kervella, P., Montargès, M., Ridgway, S. T., et al. 2014, A&A, 564, A88 [NASA ADS] [CrossRef] [EDP Sciences] [Google Scholar]
25. Kervella, P., Homan, W., Richards, A. M. S., et al. 2016, A&A, 596, A92 [NASA ADS] [CrossRef] [EDP Sciences] [Google Scholar]
26. Kippenhahn, R., & Weigert, A. 1990, Stellar Structure and Evolution (Springer-Verlag Berlin Heidelberg New York) [Google Scholar]
27. Lanza, A. F., Lebreton, Y., & Sallard, C. 2023, A&A, 674, A176 [NASA ADS] [CrossRef] [EDP Sciences] [Google Scholar]
28. Madappatt, N., De Marco, O., & Villaver, E. 2016, MNRAS, 463, 1040 [NASA ADS] [CrossRef] [Google Scholar]

29. McDonald, I., & Zijlstra, A. A. 2015, MNRAS, 448, 502 [Google Scholar]

30. Mogavero, F., & Laskar, J. 2021, A&A, 655, A1 [NASA ADS] [CrossRef] [EDP Sciences] [Google Scholar]

31. Mustill, A. J., & Villaver, E. 2012, ApJ, 761, 121 [NASA ADS] [CrossRef] [Google Scholar]

32. Nordhaus, J., Spiegel, D. S., Ibgui, L., Goodman, J., & Burrows, A. 2010, MNRAS, 408, 631 [NASA ADS] [CrossRef] [Google Scholar]

33. Ogilvie, G. I. 2014, ARA&A, 52, 171 [Google Scholar]

34. Olofsson, H., González Delgado, D., Kerschbaum, F., & Schöier, F. L. 2002, A&A, 391, 1053 [NASA ADS] [CrossRef] [EDP Sciences] [Google Scholar]

35. Parkosidis, A., Toonen, S., Dosopoulou, F., & Laplace, E. 2026a, A&A, 706, A79 [NASA ADS] [CrossRef] [EDP Sciences] [Google Scholar]

36. Parkosidis, A., Toonen, S., Laplace, E., & Dosopoulou, F. 2026b, A&A, 706, A357 [NASA ADS] [CrossRef] [EDP Sciences] [Google Scholar]

37. Paxton, B., Bildsten, L., Dotter, A., et al. 2011, ApJS, 192, 3 [Google Scholar]

38. Paxton, B., Cantiello, M., Arras, P., et al. 2013, ApJS, 208, 4 [Google Scholar]

39. Paxton, B., Marchant, P., Schwab, J., et al. 2015, ApJS, 220, 15 [Google Scholar]

40. Paxton, B., Schwab, J., Bauer, E. B., et al. 2018, ApJS, 234, 34 [NASA ADS] [CrossRef] [Google Scholar]

41. Paxton, B., Smolec, R., Schwab, J., et al. 2019, ApJS, 243, 10 [Google Scholar]

42. Rasio, F. A., Tout, C. A., Lubow, S. H., & Livio, M. 1996, ApJ, 470, 1187 [Google Scholar]

43. Rauer, H., Aerts, C., Cabrera, J., et al. 2025, Exp. Astron., 59, 26 [Google Scholar]

44. Reimers, D. 1975, in Problems in Stellar Atmospheres and Envelopes (New York: Springer-Verlag), 229 [Google Scholar]

45. Rybicki, K. R., & Denis, C. 2001, Icarus, 151, 130 [NASA ADS] [CrossRef] [Google Scholar]

46. Sackmann, I.-J., Boothroyd, A. I., & Kraemer, K. E. 1993, ApJ, 418, 457 [Google Scholar]

47. Saladino, M. I., Pols, O. R., & Abate, C. 2019, A&A, 626, A68 [NASA ADS] [CrossRef] [EDP Sciences] [Google Scholar]

48. Schröder, K.-P., & Smith, R. C. 2008, MNRAS, 386, 155 [CrossRef] [Google Scholar]

49. Uttenthaler, S. 2024, A&A, 692, A224 [NASA ADS] [CrossRef] [EDP Sciences] [Google Scholar]

50. Vassiliadis, E., & Wood, P. R. 1993, ApJ, 413, 641 [Google Scholar]

51. Zahn, J. P. 1966, Ann. Astrophys., 29, 313 [NASA ADS] [Google Scholar]

52. Zahn, J. P. 1977, A&A, 57, 383 [Google Scholar]

53. Zhang, K., Zang, W., El-Badry, K., et al. 2024, Nat. Astron., 8, 1575 [Google Scholar]

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[1] Although the simulations of Duguid et al. (2020) are currently the best in the state of the art, they still don't perfectly represent true stellar conditions. They only compute values of the Rayleigh number up to 10^3, where although their results are relatively robust against the Rayleigh

number in their regime, deviations to the prescription when adopting more realistic Rayleigh numbers (and Prandtl numbers; for the current Sun Ra $\in [10^{21}, 10^{24}]$ and Pr $\in [10^{-7}, 10^{-3}]$ Hanasoge et al. 2016) would not be unexpected.

Appendix A: Orbital evolution equations for eccentricity

The equations for the evolution of the eccentricity computed in Esseldeurs et al. (2026) are given by

$$\frac{\mathrm{d}e^2}{\mathrm{d}t} = \frac{\mathrm{d}e^2}{\mathrm{d}t}\Big|_{\mathrm{wind}} + \frac{\mathrm{d}e^2}{\mathrm{d}t}\Big|_{\mathrm{tides}} \tag{A.1}$$

where both the contributions of the stellar wind and the tides are given by

$$\left\langle \frac{\mathrm{d}e^2}{\mathrm{d}t} \right\rangle\Big|_{\mathrm{wind}} = -4e^2 \frac{\dot{m}_1}{m_1} \langle\beta\rangle \left(\frac{1}{q} - \eta\frac{1+q}{q} - \frac{1}{2}\frac{1}{1+q}\right) \tag{A.2}$$

where $\langle\rangle$ denotes orbit-averaged quantities, $\langle\beta\rangle = \beta_{\mathrm{c}} \frac{1}{\sqrt{1-e^2}}$, and

$$\frac{\mathrm{d}e^2}{\mathrm{d}t}\Big|_{\mathrm{tides}} = -\frac{5}{4\pi}\frac{m_2}{m_1}\left(\frac{R_\star}{a}\right)^5 \sqrt{1-e^2}\,\Omega_o \quad \sum_{m,n} \left(n\sqrt{1-e^2} - m\right) \times |A_{2,m,n}|^2 \, I(k_n^{2,m}) \tag{A.3}$$

Appendix B: Kippenhahn diagram

For the $\eta_{\text{Blöcker}} = 0.05$ model the Kippenhahn diagram is shown in Fig. B.1, indicating the different stellar evolutionary phases, as well as the convective and radiative regions. The stellar age is visualised as the time until the end of the simulation, that is, the time until the WD is cool enough to produce a luminosity of $L = 10^{-1}$ $\mathrm{L}_\odot$. During the main sequence, the Sun has a convective envelope and a radiative core. During the RGB phase, the convective envelope deepens and the star expands. During the helium flash, the star contracts and the convective envelope recedes, creating a radiative shell around a convective core (a three-layer structure). During the AGB phase, the star expands again and the convective envelope deepens again. Finally when the star sheds its envelope, it contracts again and becomes a white dwarf.

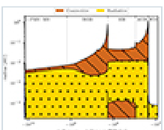


Fig. B.1.

Kippenhahn diagram of the evolution of the Sun. The brown hatched regions indicate convective zones, while the yellow dotted regions indicate radiative zones. The stellar age (x-axis) is visualised as the time until the end of the simulation in logarithmic scale to better distinguish the different evolutionary phases.

Appendix C: Previous tidal dissipation prescriptions

Following Zahn (1966, 1977), Hut (1981), Hurley et al. (2002), Mustill & Villaver (2012), and rewriting the equations in terms of the tidal Love numbers, the changes through tides can be written as

$$I(k_2^{2,2})_{\rm eq} = \frac{1}{27}\frac{1}{t_{\rm conv}}\frac{m_{\rm env}}{Gm_1^2}R_\star^3\,\omega_t F(\omega_t)\,, \tag{C.1}$$

where $t_{\rm conv}$ is the convective turnover timescale, $m_{\rm env}$ is the mass of the convective envelope, ω_t is the tidal frequency, and $F(\omega_t)$ is a frequency-dependent factor that accounts for the interaction between turbulent convection and tidal flows (see e.g. Duguid et al. 2020). This relation assumes that the tidal dissipation is dominated by the equilibrium tide, as well as parametrisations for stellar properties. For the frequency-dependent factor, we use the prescription of Mustill & Villaver (2012), where

$$t_{\rm conv} = \left(\frac{m_{\rm env}R_{\rm env}^2}{\eta_F L}\right)^{1/3}, \quad F(\omega_t) = f' \min\left(1, \left|\frac{2\pi}{\omega_t c_F t_{\rm conv}}\right|^{\gamma_F}\right), \tag{C.2}$$

where η_F, f, c_F, and γ_F are parameters of order unity. We adopt their parameters of $\eta_F = 3$, $f = 9/2$, $c_F = 1$, and $\gamma_F = 2$.

A first difference with the prescriptions of Esseldeurs et al. (2024) is that in Esseldeurs et al. (2024) the dynamical tide associated with progressive internal gravity waves is also included, while in the prescriptions of Mustill & Villaver (2012) only the equilibrium tide is included. However, this dynamical tide is expected to be subdominant in the giant phases (see Beck et al. 2018, 2022, 2024; Esseldeurs et al. 2024; Dewberry & Wu 2025; Esseldeurs et al. 2026). A second difference comes from the different parametrisations of the equilibrium tide, where the prescriptions of Zahn (1966) evaluate the convective turnover timescale as one value for the entire convective envelope, while the prescriptions of Esseldeurs et al. (2024) evaluate the convective turnover timescale locally at each radius in the convective envelope, which leads to a more accurate evaluation of the tidal dissipation. The bulk of the convective envelope however has a relatively uniform convective turnover timescale, so the difference is not expected to be large. Lastly, the prescriptions of Esseldeurs et al. (2024) are calibrated on numerical simulations of convection of Duguid et al. (2020)[1], while previous studies like Mustill & Villaver (2012) use the Goldreich & Keeley (1977)$\gamma_F = 2$ fast-tide scaling. The Earth-Sun system hovers around the transition between the fast and slow tide regime ($\omega_t t_{\rm conv} \approx 1$), exactly where the difference between the two prescriptions is the largest and can change up to an order of magnitude. This makes the choice of prescription for the interaction between convective turbulence and tidal flows crucial for the orbital evolution of the Earth.

Appendix D: Mean motion resonances

In our modeling of the orbital evolution of the inner Solar System, we do not take into account the effect of planet-planet interactions on the orbital evolution. However, as the star evolves and the planets move outward, they can cross mean motion resonances with other planets, which can lead to changes in the eccentricity and inclination of the planets, and thus affect their orbital evolution. To check if this is the case for Earth, we show in Fig. D.1 the period ratio of the Earth with respect to Venus and Mars during the evolved phases of the Sun. It can be seen that Earth does not cross any first-order mean motion resonance with Venus or Mars during the evolved phases of the Sun, however Venus-Earth does cross the 5:3 second-order mean motion resonance during the RGB phase. This means that the effect of planet-planet interactions on the orbital evolution of the Earth could be important during the RGB phase, but is likely negligible during the AGB phase.

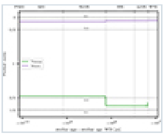


Fig. D.1.

Period ratio of the Earth with respect to the closest planets, where green represents Venus and purple represents Mars.

Appendix E: The mass-loss rate of L_2 Pup

The AGB star L_2 Pup is an important observational proxy for the Sun during the AGB phase, as its predicted initial mass is close to that of the Sun (0.98 $M_\odot$; Kervella et al. 2016). There are two different estimates of the mass-loss rate of L_2 Pup: one based on the dust emission, which gives a dust mass-loss rate of $\dot{m}_{1,d} = 2.5 \times 10^{-9}$ $M_\odot$ yr^{-1} and a gas-to-dust ratio of 400, corresponds to a total mass-loss rate of $\dot{m}_1 = 1 \times 10^{-6}$ $M_\odot$ yr^{-1} (Haworth et al. 2018); and one based on the CO emission, which gives a much lower mass-loss rate of $\dot{m}_1 = 1.2 \times 10^{-8}$ $M_\odot$ yr^{-1} (Hoai et al. 2022). Both estimates take into account the effect of the disk around L_2 Pup and derive mass-loss rates similar to previous estimates that did not account for the disk (see, e.g., Olofsson et al. 2002; Bedding et al. 2002; Jura et al. 2002; Danilovich et al. 2015). The estimate based on the dust emission likely best traces the motion of material close to the star (i.e., within the disk), while the estimate based on the CO emission likely best traces the motion of material in the outer regions of the circumstellar environment. Both methods have their uncertainties; however, it is less clear whether the dust mass-loss rate can be directly converted into a gas mass-loss rate using a fixed gas-to-dust ratio, although this is unlikely to change the inferred value by more than an order of magnitude. Using a radius of 123 $R_\odot$ for L_2 Pup (Kervella et al. 2014), a mass of 0.66 $M_\odot$ (Kervella et al. 2016), and a luminosity of 1480 $L_\odot$ (Uttenthaler 2024), the dust-based mass-loss rate corresponds to $\eta_{\text{Blöcker}} = 2.5$, while the CO-based mass-loss rate corresponds to $\eta_{\text{Blöcker}} = 0.03$. Given the large difference between the two estimates, it remains unclear which one best represents the true AGB mass-loss rate of the Sun. For this reason, we adopt $\eta_{\text{Blöcker}} = 0.05$ as a reference value, as it lies within the uncertainty range of the CO-based estimate towards the dust-based estimate.

All Figures

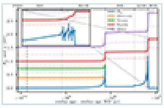


Fig. 1.

Orbital evolution (solid line, and the corresponding Solar Roche lobe radius, dotted line) of the inner Solar System during the evolved phases of the Sun. The different lines correspond to different planets (Mercury in orange, Venus in green, Earth in red, and Mars in purple). The current age of the Sun is indicated by the vertical dashed line.

In the text

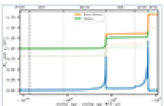


Fig. 2.

Orbital evolution of the Earth during the evolved phases of the Sun for different tidal dissipation prescriptions. In orange, we plot Esseldeurs et al. (2024) and in green, we plot Mustill & Villaver (2012) (based on Zahn 1966; see Appendix C).

In the text

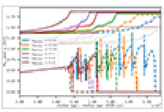
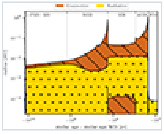


Fig. 3.

Orbital evolution of the Earth during the evolved phases of the Sun for different AGB mass-loss rates. The different lines correspond to different values for $\eta_{\text{Blöcker}}$. The dashed lines correspond to the Solar radius, and the dotted lines correspond to the Solar Roche lobe radius. After the radius of the Sun exceeds the Roche lobe radius, the solid line for the orbital distance is changed to a dashed line to highlight the uncertainty in the model after this point (see text).

[In the text](#)

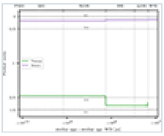
Fig. B.1.



Kippenhahn diagram of the evolution of the Sun. The brown hatched regions indicate convective zones, while the yellow dotted regions indicate radiative zones. The stellar age (x-axis) is visualised as the time until the end of the simulation in logarithmic scale to better distinguish the different evolutionary phases.

In the text

Fig. D.1.



Period ratio of the Earth with respect to the closest planets, where green represents Venus and purple represents Mars.

In the text

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